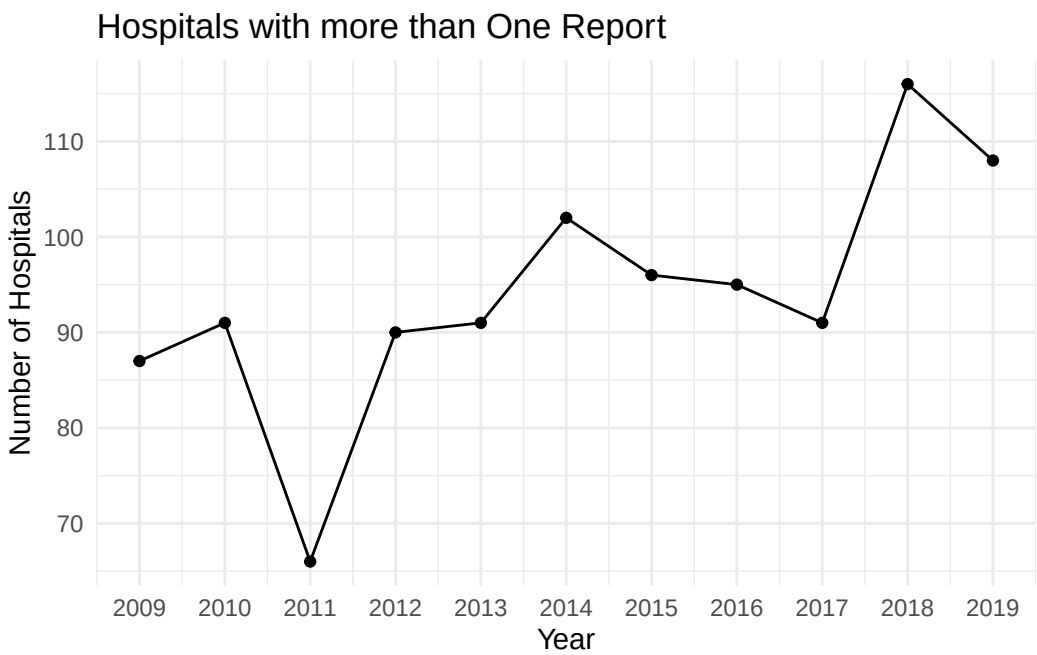
Homework 1 submission 1

Question 1:

How many hospitals filed more than one report in the same year? Show answer as a line graph of the number of hospitals over time.



Question 2

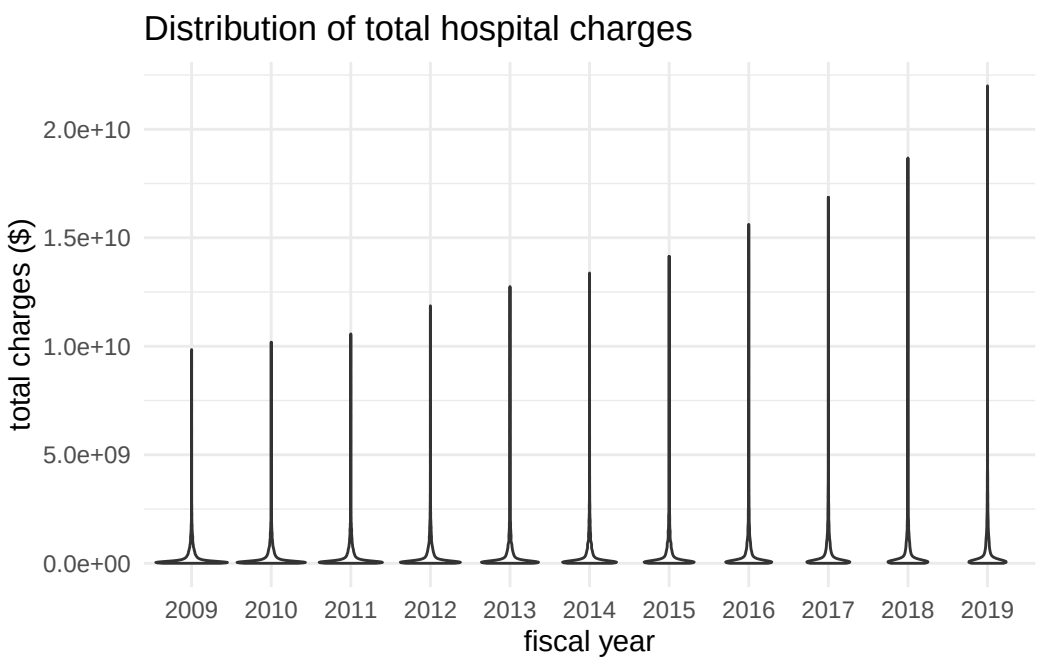
after removing/combining multiple reports, how many unique hospital IDs (Medicare provider numbers) exist in the data?

```
[1] "Total unique hospitas IDs 6897"
```

```
# A tibble: 11 x 2
    year hospitals
   <dbl>     <int>
 1  2009      6100
 2  2010      6103
 3  2011      6097
 4  2012      6140
 5  2013      6066
 6  2014      6064
 7  2015      6051
 8  2016      6091
 9  2017      6083
10  2018      6042
11  2019      6035
```

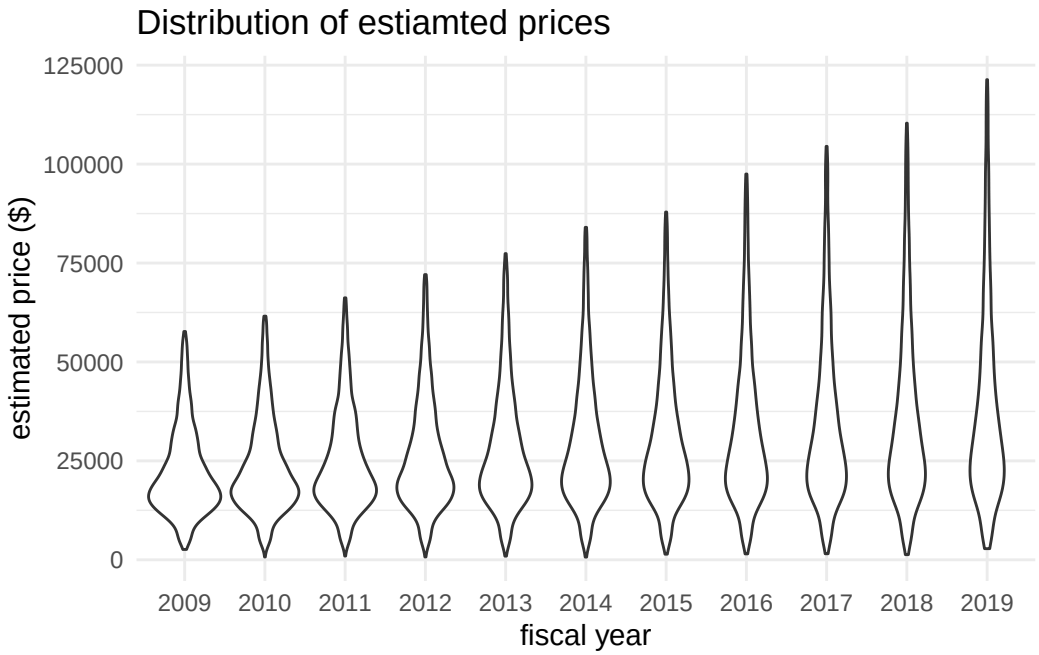
Question 3

What is the distribution of total charges (tot_charges) in each year? Show results in a violin plot with charges on the y-axis and years on the x-axis



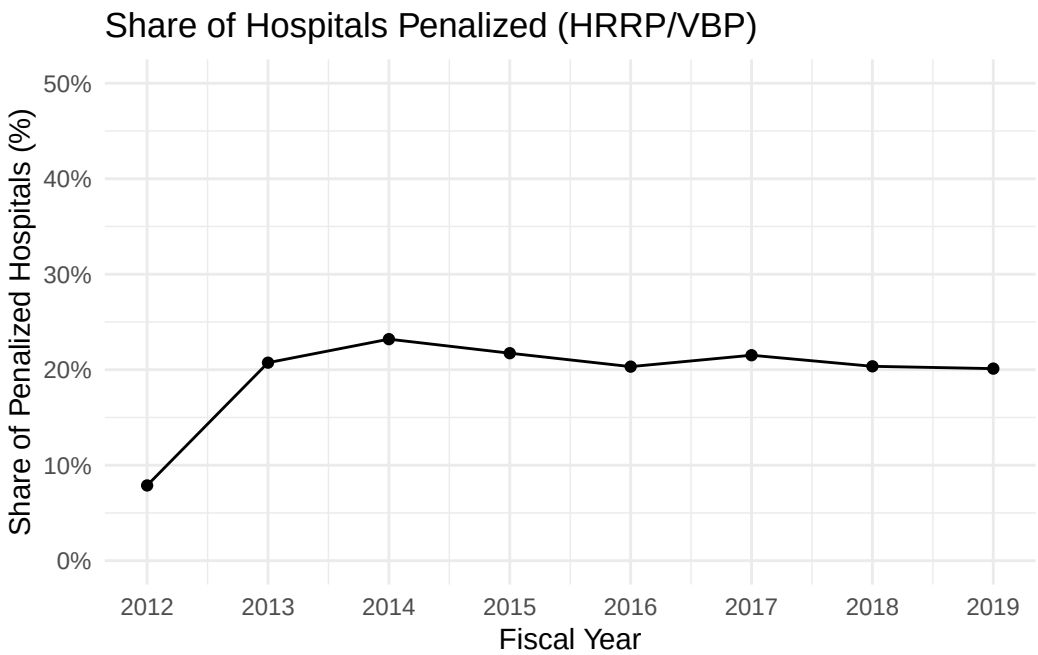
Question 4

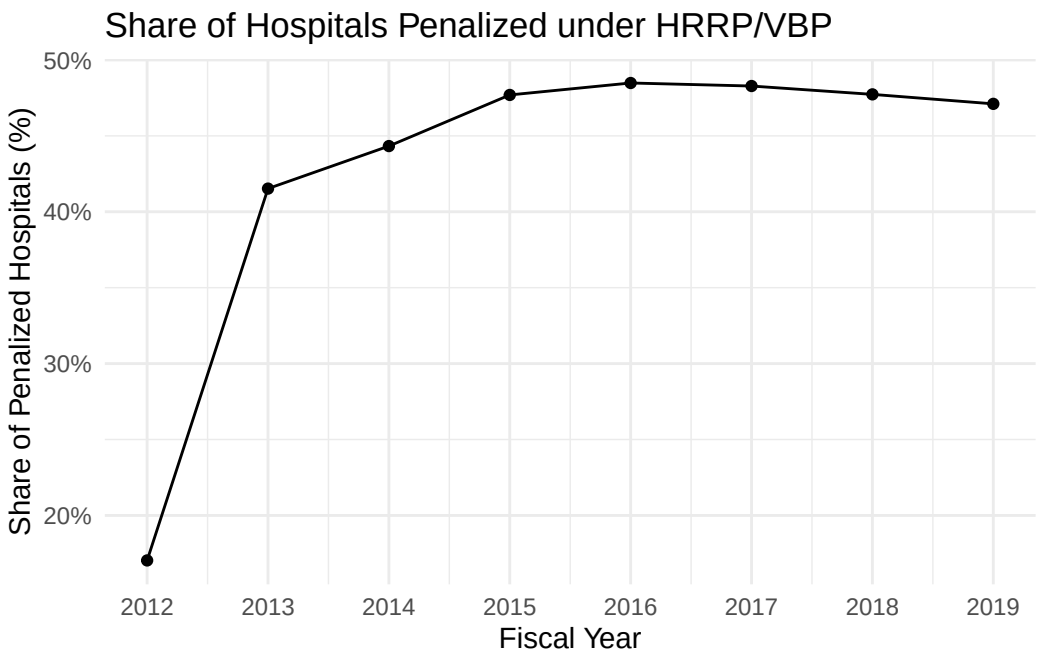
what is the distribution of estimated prices in each year? Present results with a violin plot (be sure to do something about outliers and/or negative prices in the data)



Question 5

What share of hospitals penalized under HRRP/VBP? Provide a graph showing the share of penalized hospitals from 2012-2019.





Question 6

Provide a summary of OLS estimates of the effect of net penalties on price changes. Present your results in a table with three different specifications: 1) a "baseline" specification using

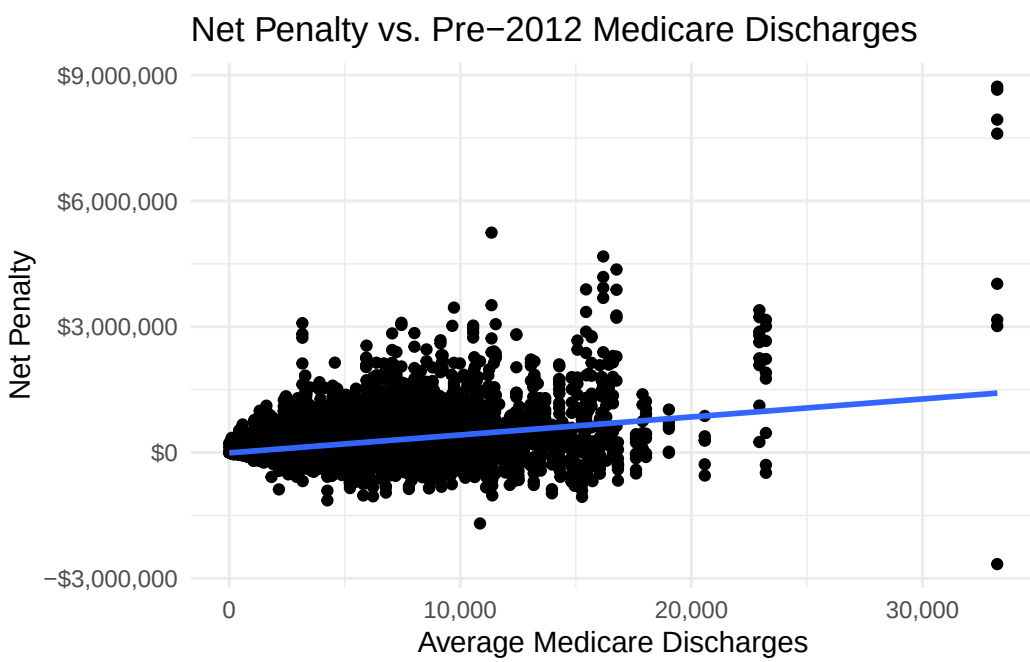
only net penalty as a covariate; 2) "baseline" specification plus the pre-penalty (2009-2011) mean bed size; 3) "baseline" specification plus bed size plus pre-penalty (2009-2011) average Medicaid discharges.

```
                              mod1                mod2                mod3
                          Baseline       with Bed size          Full Model
Dependent Var.:        price_change        price_change        price_change

Constant         0.3606*** (0.0697) 0.3566*** (0.0706) 0.3503*** (0.0868)
hrrp_payment     -4.15e-7 (5.13e-7) -4.29e-7 (5.15e-7) -5.16e-7 (5.53e-7)
beds                                 3.46e-5 (8.69e-5)  2.14e-5 (9.26e-5)
mcaid_discharges                                        4.11e-5 (4.73e-5)
---------------- ------------------ ------------------ ------------------
S.E. type                       IID                IID                IID
Observations                133,212            133,072            119,336
R2                          4.91e-6             6.1e-6            1.29e-5
Adj. R2                     -2.6e-6           -8.93e-6           -1.23e-5
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 7

Provide a scatterplot of net penalty against pre-2012 Medicare discharges.



Question 8

Provide a summary of the first stage and reduced-form results using pre-penalty Medicare discharges as an instrument for net penalties. Present your results in a table with three different specifications as in Question 6.

```
                                     fs1                     fs2
                                Baseline          with Bed Size
Dependent Var.:                net_penalty            net_penalty

Constant             -13,403.5*** (2,797.6) -13,200.1*** (2,798.1)
pre_mcare_discharges      44.72*** (0.6074)      44.92*** (0.6112)
pre_bed_size                                      -3.729** (1.287)
pre_mcaid_discharges
-------------------- ---------------------- ----------------------
S.E. type                               IID                    IID
Observations                         24,410                 24,410
R2                                  0.18171                0.18199
Adj. R2                             0.18168                0.18192

                                     fs3
                            with Medicaid
Dependent Var.:               net_penalty

Constant             -8,126.5** (2,844.8)
pre_mcare_discharges    50.92*** (0.7171)
pre_bed_size              -3.147* (1.286)
pre_mcaid_discharges    -17.26*** (1.089)
-------------------- --------------------
S.E. type                             IID
Observations                       24,171
R2                                0.19004
Adj. R2                           0.18994
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                                   rf1                 rf2                 rf3
                              Baseline       with Bed Size       with Medicaid
Dependent Var.:            price_change        price_change        price_change

Constant                 0.1247* (0.0614)    0.1240* (0.0614)     0.0956 (0.0663)
pre_mcare_discharges 0.0002*** (1.8e-5) 0.0002*** (1.82e-5) 0.0001*** (2.32e-5)
```

```
pre_bed_size                                1.9e-5 (4.91e-5)   1.13e-5 (5.05e-5)
pre_mcaid_discharges                                            0.0001*** (3.65e-5)
-------------------- ------------------ ------------------- -------------------
S.E. type                           IID                 IID                 IID
Observations                     51,402              51,402              48,303
R2                              0.00160             0.00161             0.00187
Adj. R2                         0.00158             0.00157             0.00181
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 9

Provide a summary of IV estimates of the effect of net penalties on price changes. Again present your results in a table with the three different specifications as in Questions 6 and 8.

```
                                     iv1                 iv2
                                Baseline       with Bed Size
Dependent Var.:             price_change        price_change

Constant                  0.1557** (0.0575)   0.1554** (0.0576)
net_penalty             8.86e-7** (2.93e-7) 8.82e-7** (2.93e-7)
pre_bed_size                                3.96e-6 (2.77e-5)
pre_mcaid_discharges
-------------------- ------------------ ------------------
S.E. type                           IID                 IID
R2                              -0.00298            -0.00295
Observations                      24,410              24,410

                                     iv3
                                Medicaid
Dependent Var.:             price_change

Constant                  0.1576** (0.0604)
net_penalty             8.55e-7** (3.05e-7)
pre_bed_size              3.75e-6 (2.78e-5)
pre_mcaid_discharges      2.19e-6 (2.14e-5)
-------------------- ------------------
S.E. type                           IID
R2                              -0.00281
Observations                      24,171
---
```

```
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 10

Briefly explain the "Local" ATE in the context of your estimates. How might a local effect differ from an overall ATE in this setting?

The local ATE refers to the effect of the penalties on price changes for the hospitals whose penalty status was changed by the pre-2012 medicare discharges. The ATE would be the impact of penalties on any hospital chosen randomly, local ATE is weighted towards larger Medicare-dependent hospitals.